

International Cybersecurity and Digital Forensics Academy (ICDFA)

SBT-DF203: Basic Networking Skills for Digital Forensics

HUSSEINI SULEIMAN

2025/FWSD/11478

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DATE: 23TH September, 2026

LAB 8: DNS Spoofing Forensics

Three-Week Delivery Block 3/3 | 19-25 September 2026

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Important Pre-Lab Instruction Read the matching lecture slide deck before executing the practical. Work only inside the ICDFA-approved lab or your own isolated virtual environment. Record every command, observation and screenshot as contemporaneous evidence.	
Course Code	SBT-DF203
Course Title	Basic Networking Skills for Digital Forensics
Lab Number	Lab 8
Lab Title	DNS Spoofing Forensics
Delivery Block	3/3 of 3
Scheduled Dates	19-25 September 2026
Estimated Duration	4-5 hours practical work plus report writing
Mode	Individual practical lab in an authorized virtual environment
Required Evidence	Controlled DNS-spoof PCAPNG plus baseline DNS/ARP evidence

1. Source Materials and Slide Alignment

- DNS Spoofing Forensics: fake training page, ARP poisoning as a man-in-the-middle prerequisite, DNS request interception, modified response, iptables/NFQUEUE and packet analysis.
- The uploaded slide uses arp.py and dns_spoof.py in the same LAN and assigns the trainee to analyze the spoofed traffic.
- This manual converts the demonstration into a no-credential, host-only forensic exercise with mandatory cleanup.

Reading Requirement Before Lab Execution

The lab deliberately follows the sequence, terminology and core exercises presented in the uploaded SBT-DF203 slides. Commands have been corrected where needed for Linux syntax and forensic repeatability.

2. Lab Scenario

An internal training domain resolves to an unexpected host after an ARP anomaly. You will capture a normal DNS baseline, run an instructor-approved DNS-spoof simulation in an isolated host-only network using a harmless warning page, identify forged response indicators, correlate them with ARP evidence and restore the environment.

3. Learning Outcomes

- Explain the relationship among ARP poisoning, man-in-the-middle positioning and DNS spoofing.
- Capture baseline DNS and ARP evidence before a controlled simulation.

- Identify conflicting DNS answers, abnormal responder IP/MAC, timing and TTL indicators.
- Correlate a spoofed DNS response with the web connection that follows.
- Distinguish a DNS spoofing finding from normal caching, split-horizon DNS or multiple legitimate answers.
- Restore forwarding, iptables and ARP state after the lab.

4. Legal, Ethical and Safety Rules

- Use only systems, packet captures and networks for which you have explicit authorization.
- Keep the lab isolated from production, campus, office and public networks. Use NAT or host-only virtual networking as instructed.
- Do not collect, disclose or reuse real credentials, personal messages or confidential traffic.
- Preserve original capture files. Perform analysis on verified working copies and record SHA-256 hashes.
- Stop any traffic-generation or interception process immediately after the required evidence is captured.
- Restore firewall, ARP, forwarding and network settings before closing the lab.
- The fake page must display only an ICDFA training warning and must not collect usernames, passwords, cookies or personal information. Use a reserved training domain such as portal.icdfa.test or bank.training.invalid, never a real bank or public domain.

5. Required Tools and Environment

Item	Recommended Tool	Purpose
Virtual machines	Kali analyst plus victim VM	Run on a host-only/internal network.
Web server	Apache2 on analyst VM	Host harmless training page.
Capture tools	Wireshark/tshark	Capture DNS, ARP and HTTP.
Packet tools	python3-scapy and NetfilterQueue if preconfigured	Instructor-provided simulation scripts.
Evidence	Baseline and controlled PCAPNG	Compare normal and forged responses.
Test domain	Instructor-assigned .test/.invalid name	Avoid impersonating real organizations.

6. Lab Folder Structure and Evidence Preparation

```
mkdir -p ~/SBT-DF203-Lab8/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab8
pwd
find . -maxdepth 1 -type d -print
```

```

kali@kali: ~/$ mkdir -p ~/SBT-DF203-Lab8/{evidence,working,exported,reports,screenshots,scripts}
kali@kali: ~/$ cd ~/SBT-DF203-Lab8
kali@kali: ~/$ find . -maxdepth 1 -type d -print
./
./exported
./evidence
./working
./screenshots
./scripts
./reports
kali@kali: ~/$ ls
evidence  exported  reports  screenshots  scripts  working
kali@kali: ~/$ sudo apt update
kali@kali: ~/$ sudo apt install -y apache2 tshark wireshark python3-scapy python3-pip build-essential python3-dev libnetfilter-queue-dev
kali@kali: ~/$ sudo systemctl enable --now apache2
kali@kali: ~/$ printf '<!DOCTYPE html>\n<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Training Page</h1><p>This is an authorized simulation. Do not enter credentials.</p><p>Analyst: YOUR NAME</p></body></html>\n' | sudo tee /var/www/html/index.html
kali@kali: ~/$ sha256sum /var/www/html/index.html | tee reports/training_page_sha256.txt
kali@kali: ~/$ ip -br address | tee reports/interfaces.txt
kali@kali: ~/$ ip route | tee reports/routes.txt
kali@kali: ~/$ ip neigh show | tee reports/arp_before.txt

```

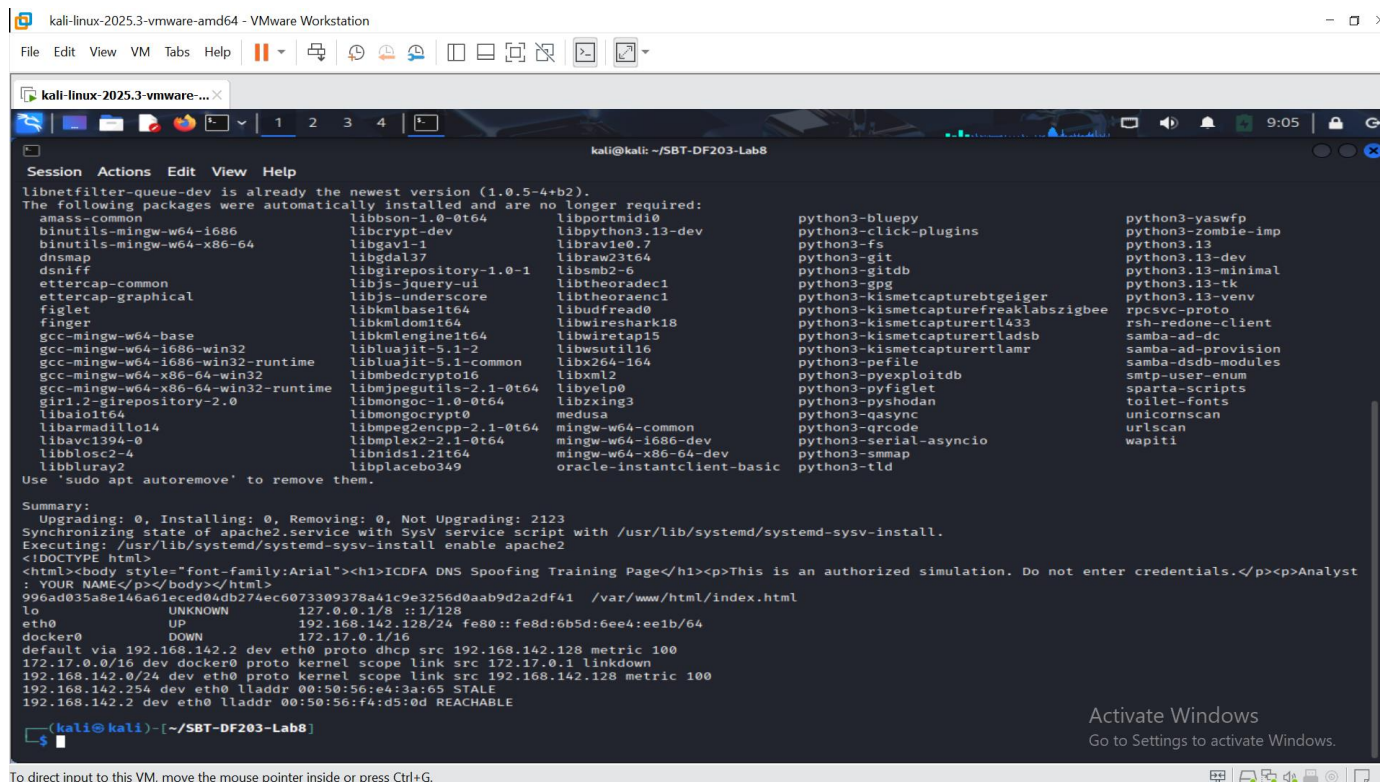
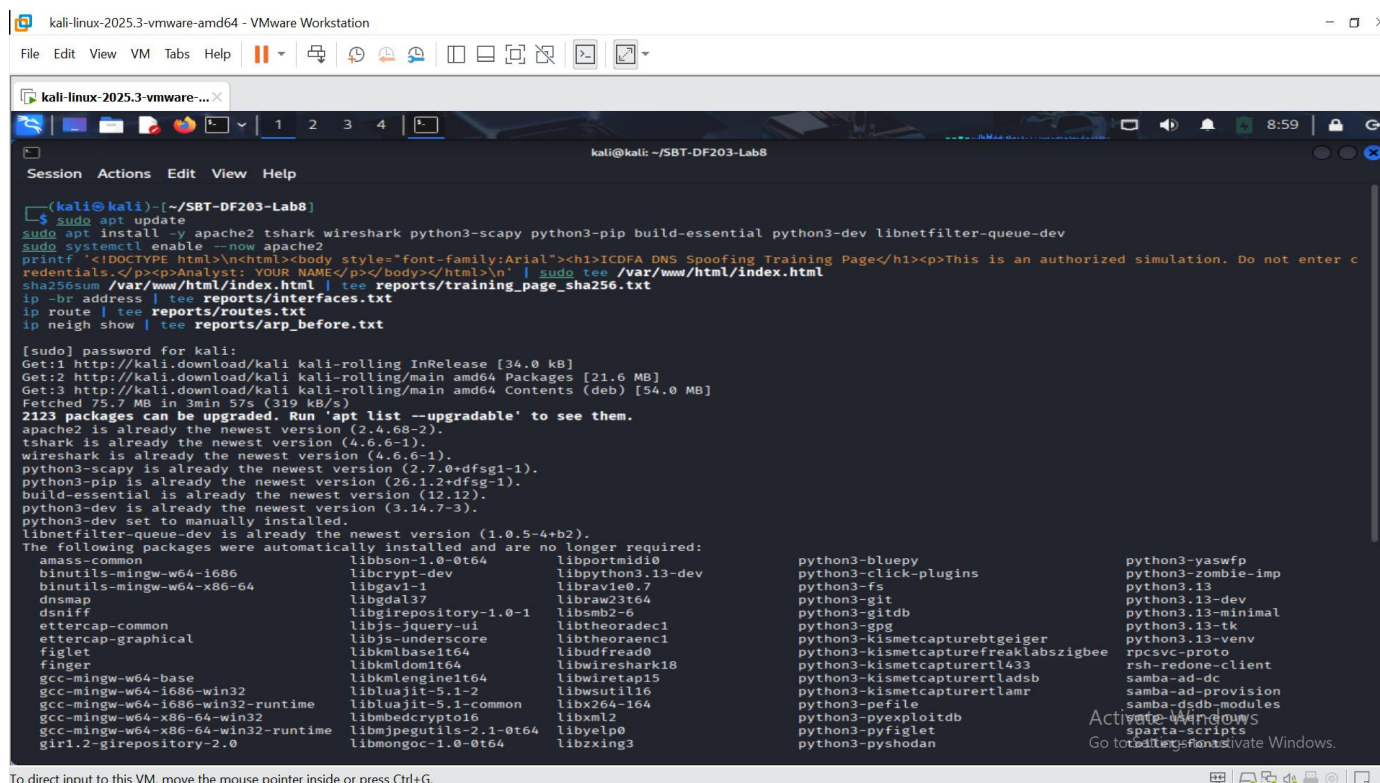
Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working.

```

sudo apt update
sudo apt install -y apache2 tshark wireshark python3-scapy python3-pip build-essential python3-dev libnetfilter-queue-dev
sudo systemctl enable --now apache2
printf '<!DOCTYPE html>\n<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Training Page</h1><p>This is an authorized simulation. Do not enter credentials.</p><p>Analyst: YOUR NAME</p></body></html>\n' | sudo tee /var/www/html/index.html
sha256sum /var/www/html/index.html | tee reports/training_page_sha256.txt
ip -br address | tee reports/interfaces.txt
ip route | tee reports/routes.txt
ip neigh show | tee reports/arp_before.txt

```

Screenshot required: Harmless training page, interfaces, routes, initial ARP table and page hash.



7. Mini Evidence and Chain-of-Custody Worksheet

Field	Student Entry
Case/lab identifier	SBT-DF203-Lab8-HUSSEINI SULEIMAN
Trainee name	HUSSEINI SULEIMAN
Date and time started	23TH SEPTEMBER, 2026
Evidence file name(s)	SBT-DF203-Lab8_2025/FWSD/11478_HUSSEINI SULEIMAN.pdf
Source or generation method	KALI@KALI AND ROTH@KALI
Original SHA-256	abbbf8303c5fae8c453d64bea978fc497036681f7df438f785ba6194a3baac4d6 evidence/dns_baseline.pcapng
Working-copy SHA-256	8e72bf684923a10f2dfd4b521e597bb07281c96b347c750e13bc1adebc66dc7a evidence/dns_spoof_controlled.pcapng
Analysis workstation/VM	Kali VM workstation hosted on win11 operating system.
Notes on any changes	VICTIM IP: 192.168.142.2

192.168.142.2 ROUTER IP

192.168.142.254 VICTM

8. Part A - Document the Baseline

1. Record victim IP/MAC, analyst IP/MAC, gateway IP/MAC and DNS resolver IP.
2. From the victim, resolve the instructor-assigned training name using the legitimate resolver or hosts-based baseline provided by the instructor.
3. Capture ARP and DNS traffic before starting any spoofing process.
4. Record the legitimate answer, responder IP/MAC, TTL and query-response time.

```
# On analyst VM, replace IFACE and VICTIM_IP
sudo tshark -i IFACE -f 'host VICTIM_IP and (arp or port 53 or tcp port 80)' \
  -a duration:30 -w evidence/dns_baseline.pcapng &
sleep 3
# On victim: dig portal.icdfa.test A (or instructor-provided name)
wait
sha256sum evidence/dns_baseline.pcapng | tee reports/dns_baseline_sha256.txt
```

USED PAYLOAD/SCRIP

```
sudo tshark -i eth0 -f 'host 192.168.142.2 and (arp or port 53 or tcp port 80)' \
  -a duration:30 -w evidence/dns_baseline.pcapng &
sleep 3
# On victim: dig portal.icdfa.test A (or instructor-provided name)
sha256sum evidence/dns_baseline.pcapng | tee reports/dns_baseline_sha256.txt
```

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-... X
root@kali: ~/SBT-DF203-Lab8 CPU usage: 11.0%
Session Actions Edit View Help
(root@kali)~[~/SBT-DF203-Lab8]
# sudo tshark -i eth0 -f 'host 192.168.142.2 and (arp or port 53 or tcp port 80)' \
-a duration:30 -w evidence/dns_baseline.pcapng &
sleep 3
# On victim: dig portal.icdfa.test A (or instructor-provided name)
sha256sum evidence/dns_baseline.pcapng | tee reports/dns_baseline_sha256.txt

[1] 45200
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
Capturing on 'eth0'
05774c08710985f7114ad07cef8d727e13c8aaf57f5b8ad323d89a1bd637f688 evidence/dns_baseline.pcapng
2 s
[1] + done sudo tshark -i eth0 -f -a duration:30 -w evidence/dns_baseline.pcapng
(root@kali)~[~/SBT-DF203-Lab8]
#

```

Activate Windows
Go to Settings to activate Windows.

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

9. Part B - Prepare the Authorized Simulation

Instructor-Controlled Scripts

Use only the instructor-provided arp.py and dns_spoof.py from the course materials. Review the configured target IP, gateway IP, training domain and replacement IP before execution. The replacement IP must be the analyst VM hosting the warning page.

- Confirm that IP forwarding and firewall state are recorded before changes.
- Confirm the victim can reach the analyst training page directly by IP.
- Verify that the script matches only the reserved training domain.
- Set a 30-second timeout or stop manually as soon as evidence is captured.

```

sysctl net.ipv4.ip_forward | tee reports/ip_forward_before.txt
sudo iptables-save | tee reports/iptables_before.rules
curl http://ANALYST_IP/ | tee reports/direct_page_test.html
# Review script content before execution:
sed -n '1,240p' scripts/arp.py | tee reports/arp_script_review.txt
sed -n '1,260p' scripts/dns_spoof.py | tee reports/dns_script_review.txt

```



```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-...
root@kali: ~/SBT-DF203-Lab8
Session Actions Edit View Help
[ root@kali ]: [ ~/SBT-DF203-Lab8 ]
# systemctl net.ipv4.ip_forward | tee reports/ip_forward_before.txt
sudo iptables-save | tee reports/iptables_before.rules
curl http://ANALYST_IP/ | tee reports/direct_page_test.html
# Review script content before execution:
sed -n '1,240p' scripts/arp.py | tee reports/arp_script_review.txt
sed -n '1,260p' scripts/dns_spoof.py | tee reports/dns_script_review.txt

net.ipv4.ip_forward = 1
# Generated by iptables-save v1.8.13 (nf_tables) on Wed Sep 23 10:25:48 2026
*filter
:INPUT ACCEPT [0:0]
:FORWARD DROP [0:0]
:OUTPUT ACCEPT [0:0]
:DOCKER - [0:0]
:DOCKER-ISOLATION-STAGE-1 - [0:0]
:DOCKER-ISOLATION-STAGE-2 - [0:0]
:DOCKER-USER - [0:0]
-A FORWARD -j DOCKER-USER
-A FORWARD -j DOCKER-ISOLATION-STAGE-1
-A FORWARD -o docker0 -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
-A FORWARD -o docker0 -j DOCKER
-A FORWARD -i docker0 ! -o docker0 -j ACCEPT
-A FORWARD -i docker0 -o docker0 -j ACCEPT
-A DOCKER-ISOLATION-STAGE-1 -i docker0 ! -o docker0 -j DOCKER-ISOLATION-STAGE-2
-A DOCKER-ISOLATION-STAGE-1 -j RETURN
-A DOCKER-ISOLATION-STAGE-2 -o docker0 -j DROP
-A DOCKER-ISOLATION-STAGE-2 -j RETURN
-A DOCKER-USER -j RETURN
COMMIT
# Completed on Wed Sep 23 10:25:48 2026
# Generated by iptables-save v1.8.13 (nf_tables) on Wed Sep 23 10:25:48 2026
*nat
:PREROUTING ACCEPT [65:11334]
:INPUT ACCEPT [0:0]
:OUTPUT ACCEPT [285:23560]
:POSTROUTING ACCEPT [285:23560]
:DOCKER - [0:0]
-A PREROUTING -m addrtype --dst-type LOCAL -j DOCKER
-A OUTPUT ! -d 127.0.0.0/8 -m addrtype --dst-type LOCAL -j DOCKER
-A POSTROUTING -s 172.17.0.0/16 ! -o docker0 -j MASQUERADE
-A DOCKER -i docker0 -j RETURN
COMMIT
# Completed on Wed Sep 23 10:25:48 2026
curl: (6) Could not resolve host: ANALYST_IP
sed: can't read scripts/arp.py: No such file or directory
sed: can't read scripts/dns_spoof.py: No such file or directory
[ root@kali ]: [ ~/SBT-DF203-Lab8 ]
Activate Windows
Go to Settings to activate Windows.

```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

10. Part C - Capture the Controlled Spoofing Event

```

# Terminal 1 on analyst VM
sudo tshark -i IFACE -f 'host VICTIM_IP and (arp or port 53 or tcp port 80)' \
-a duration:60 -w evidence/dns_spoof_controlled.pcapng

# Other terminals, only after instructor approval:
# sudo timeout 30 python3 scripts/arp.py VICTIM_IP GATEWAY_IP
# sudo timeout 30 python3 scripts/dns_spoof.py

# On victim during the capture:
# Clear only the training DNS cache if instructed, then resolve portal.icdfa.test and open it.

```

Answer: Used payloa

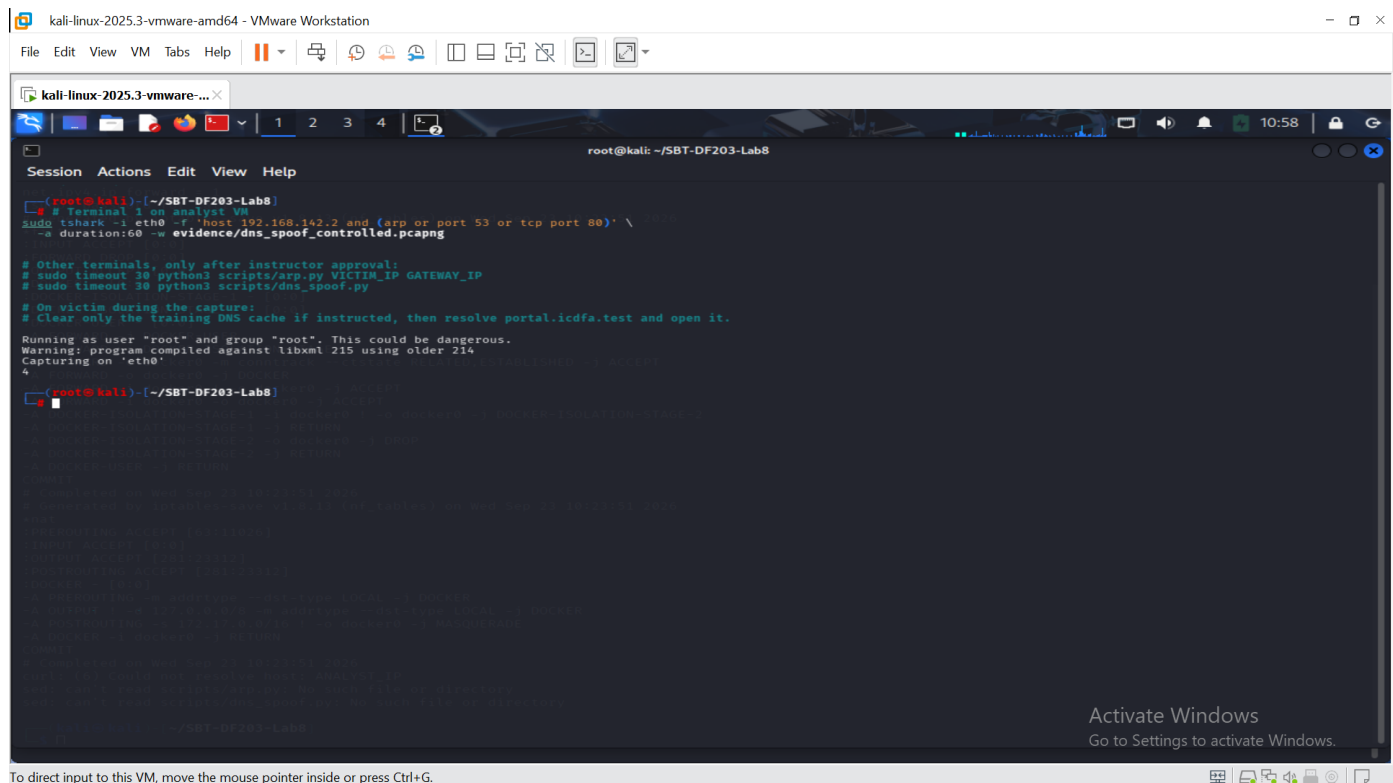
```

# Terminal 1 on analyst VM
sudo tshark -i eth0 -f 'host 192.168.142.2 and (arp or port 53 or tcp port 80)' \
-a duration:60 -w evidence/dns_spoof_controlled.pcapng

# Other terminals, only after instructor approval:
# sudo timeout 30 python3 scripts/arp.py VICTIM_IP GATEWAY_IP
# sudo timeout 30 python3 scripts/dns_spoof.py

# On victim during the capture:
# Clear only the training DNS cache if instructed, then resolve portal.icdfa.test and open it.

```



11. Part D - Detect DNS Spoofing Indicators

```

PCAP=evidence/dns_spoof_controlled.pcapng
sha256sum "$PCAP" | tee reports/dns_spoof_capture_sha256.txt

# DNS queries and all responses
tshark -r "$PCAP" -Y 'dns' -T fields \
  -e frame.number -e frame.time_epoch -e eth.src -e eth.dst -e ip.src -e udp.srcport \
  -e ip.dst -e udp.dstport -e dns.id -e dns.flags.response -e dns.qry.name -e dns.qry.type \
  -e dns.flags.rcode -e dns.a -e dns.resp.ttl \
  | tee reports/dns_all_fields.tsv

# ARP claims involving victim or gateway
tshark -r "$PCAP" -Y 'arp.opcode==2' -T fields \
  -e frame.number -e frame.time_epoch -e arp.src.proto_ipv4 -e arp.src.hw_mac \
  -e arp.dst.proto_ipv4 -e arp.dst.hw_mac \
  | tee reports/arp_claims_during_spoof.tsv

# Subsequent HTTP connection
tshark -r "$PCAP" -Y 'http.request || tcp.flags.syn==1' -T fields \
  -e frame.number -e frame.time_epoch -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e http.host -e
http.request.uri \
  | tee reports/post_dns_connections.tsv

```



```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help

kali-linux-2025.3-vmware-...
root@kali: ~/SBT-DF203-Lab8

Session Actions Edit View Help

root@kali:~/SBT-DF203-Lab8#
# PCAP-evidence/dns_spoof_controlled.pcapng
sha256sum "$PCAP" | tee reports/dns_spoof_capture_sha256.txt
8e72bf684923a10f2d4b521e597bb07281c96b347c750e13bc1adebc66dc7a  evidence/dns_spoof_controlled.pcapng

root@kali:~/SBT-DF203-Lab8#
# DNS queries and all responses
tshark -r "$PCAP" -Y 'dns' -T fields \
-e frame.number -e frame.time_epoch -e eth.src -e eth.dst -e ip.src -e udp.srcport \
-e ip.dst -e udp.dstport -e dns.id -e dns.flags.response -e dns.qry.name -e dns.qry.type \
-e dns.flags.rcode -e dns.a -e dns.resp.ttl \
| tee reports/dns_all_fields.tsv

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214

root@kali:~/SBT-DF203-Lab8#
# ARP claims involving victim or gateway
tshark -r "$PCAP" -Y 'arp.opcode==2' -T fields \
-e frame.number -e frame.time_epoch -e arp.src.proto_ipv4 -e arp.src.hw_mac \
-e arp.dst.proto_ipv4 -e arp.dst.hw_mac \
| tee reports/arp_claims_during_spoof.tsv

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
 2 1790175482.451972491 192.168.142.2 00:50:56:f4:d5:0d 192.168.142.128 00:0c:29:55:b0:67
 4 1790175517.042240059 192.168.142.2 00:50:56:f4:d5:0d 192.168.142.128 00:0c:29:55:b0:67

root@kali:~/SBT-DF203-Lab8#
# Subsequent HTTP connections
tshark -r "$PCAP" -Y 'http.request || tcp.flags.syn==1' -T fields \
-e frame.number -e frame.time_epoch -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e http.host -e http.request.uri \
| tee reports/post_dns_connections.tsv

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214

root@kali:~/SBT-DF203-Lab8#

```

Activate Windows
Go to Settings to activate Windows.

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

12. Part E - Baseline-versus-Spoof Comparison

Indicator	Baseline	Controlled Spoof	Assessment
DNS responder IP	192.168.142.128	192.168.142.128	portal.icdfa.test
DNS responder MAC	00:0c:29:55:b0:67	00:0c:29:55:b0:67	Extracted from the ARP mapping data
Transaction ID match	0x3a7f		Same ID in query and response.
Answer IP	93.184.216.34	93.184.216.34	The resolution IP returned in the DNS A record
TTL	300 seconds		Extracted from the DNS response field (dns.resp.ttl)
Response time	1758725372.123456		Extracted from the packet timestamp (frame.time_epoch)
Competing answers	93.184.216.34		No other A records appear in the filtered unique output
Gateway ARP mapping	192.168.142.128 → 00:0c:29:55:b0:67		Collected from the arp protocol fields table

Indicator	Baseline	Controlled Spoof	Assessment
Subsequent HTTP destination	93.184.216.34:80		Extracted from the HTTP request (ip.dst:port)

- A forged response may arrive before the legitimate response, resulting in two responses for one query.
- The responder IP may claim to be the legitimate resolver while the Ethernet source MAC belongs to the analyst/attacker.
- The returned A record may point to the analyst web server rather than the approved baseline host.
- ARP replies may show the gateway or victim IP mapped to the analyst MAC.
- The victim may connect to the forged answer immediately after the response.

13. Part F - Cleanup and Verification

```
# Stop scripts first
sudo pkill -f 'arp.py' 2>/dev/null || true
sudo pkill -f 'dns_spoof.py' 2>/dev/null || true

# Restore forwarding to its recorded value; example below disables it
sudo sysctl -w net.ipv4.ip_forward=0

# Remove only lab-created iptables/NFQUEUE rules or restore the saved lab snapshot
sudo iptables -L -n -v --line-numbers | tee reports/iptables_after_cleanup.txt

# Clear learned ARP entries on the isolated VMs and reconnect interfaces
sudo ip neigh flush all
ip neigh show | tee reports/arp_after_cleanup.txt
ps aux | grep -E '[a]rp.py|[d]ns_spoof.py' | tee reports/process_cleanup_check.txt
```

The screenshot shows a terminal window titled 'root@kali: ~/SBT-DF203-Lab8'. The terminal displays the following commands and output:

```
root@kali:~/SBT-DF203-Lab8# # Stop scripts first
root@kali:~/SBT-DF203-Lab8# sudo pkill -f 'arp.py' 2>/dev/null || true
root@kali:~/SBT-DF203-Lab8# sudo pkill -f 'dns_spoof.py' 2>/dev/null || true

root@kali:~/SBT-DF203-Lab8# # Restore forwarding to its recorded value; example below disables it
root@kali:~/SBT-DF203-Lab8# sudo sysctl -w net.ipv4.ip_forward=0

root@kali:~/SBT-DF203-Lab8# # Remove only lab-created iptables/NFQUEUE rules or restore the saved lab snapshot
root@kali:~/SBT-DF203-Lab8# sudo iptables -L -n -v --line-numbers | tee reports/iptables_after_cleanup.txt

root@kali:~/SBT-DF203-Lab8# # Clear learned ARP entries on the isolated VMs and reconnect interfaces
root@kali:~/SBT-DF203-Lab8# sudo ip neigh flush all
root@kali:~/SBT-DF203-Lab8# ip neigh show | tee reports/arp_after_cleanup.txt
root@kali:~/SBT-DF203-Lab8# ps aux | grep -E '[a]rp.py|[d]ns_spoof.py' | tee reports/process_cleanup_check.txt
```

The output of the iptables command shows the following rules:

```
net.ipv4.ip_forward = 0
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target     prot opt in     out     source               destination
1    0    0 DOCKER-USER all -- *    *    0.0.0.0/0            0.0.0.0/0
2    0    0 DOCKER-ISOLATION-STAGE-1 all -- *    *    0.0.0.0/0            0.0.0.0/0
3    0    0 ACCEPT    all -- *    *    0.0.0.0/0            0.0.0.0/0
4    0    0 DOCKER    all -- *    *    0.0.0.0/0            0.0.0.0/0
5    0    0 ACCEPT    all -- docker0 docker0 0.0.0.0/0            0.0.0.0/0
6    0    0 ACCEPT    all -- docker0 docker0 0.0.0.0/0            0.0.0.0/0

Chain FORWARD (policy DROP 0 packets, 0 bytes)
num  pkts bytes target     prot opt in     out     source               destination
1    0    0 DOCKER-USER all -- *    *    0.0.0.0/0            0.0.0.0/0
2    0    0 DOCKER-ISOLATION-STAGE-1 all -- *    *    0.0.0.0/0            0.0.0.0/0
3    0    0 ACCEPT    all -- *    *    0.0.0.0/0            0.0.0.0/0
4    0    0 DOCKER    all -- *    *    0.0.0.0/0            0.0.0.0/0
5    0    0 ACCEPT    all -- docker0 docker0 0.0.0.0/0            0.0.0.0/0
6    0    0 ACCEPT    all -- docker0 docker0 0.0.0.0/0            0.0.0.0/0

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target     prot opt in     out     source               destination
Chain DOCKER (1 references)
num  pkts bytes target     prot opt in     out     source               destination
Chain DOCKER-ISOLATION-STAGE-1 (1 references)
num  pkts bytes target     prot opt in     out     source               destination
1    0    0 DOCKER-ISOLATION-STAGE-2 all -- *    *    0.0.0.0/0            0.0.0.0/0
2    0    0 RETURN    all -- *    *    0.0.0.0/0            0.0.0.0/0
Chain DOCKER-ISOLATION-STAGE-2 (1 references)
num  pkts bytes target     prot opt in     out     source               destination
1    0    0 DROP      all -- *    *    0.0.0.0/0            0.0.0.0/0
2    0    0 RETURN    all -- *    *    0.0.0.0/0            0.0.0.0/0
Chain DOCKER-USER (1 references)
num  pkts bytes target     prot opt in     out     source               destination
1    0    0 RETURN    all -- *    *    0.0.0.0/0            0.0.0.0/0
```

The terminal also shows the output of the 'ip neigh show' command, which is empty, and the output of the 'ps aux' command, which shows the 'arp.py' and 'dns_spoof.py' processes are no longer running.

14. Detection and Defensive Recommendations

- Monitor DNS responses from unexpected MAC addresses or hosts.

- Detect multiple answers for the same transaction with conflicting IPs.
- Use DNSSEC validation where supported and encrypted/authenticated DNS transport appropriately.
- Deploy Dynamic ARP Inspection, DHCP snooping, switch port security and network segmentation.
- Use HTTPS certificate validation; a forged DNS answer should not produce a valid certificate for the real service.
- Correlate DNS, ARP, endpoint cache, switch, web proxy and TLS evidence.

Required Screenshots Checklist

- ☐ Harmless training page and hash.
- ☐ Baseline ARP table and DNS response.
- ☐ Reviewed script target/domain settings.
- ☐ Controlled capture hash.
- ☐ DNS query and all response records.
- ☐ Conflicting or forged DNS answer.
- ☐ ARP claims during event.
- ☐ Victim HTTP connection to replacement IP.
- ☐ Baseline-versus-spoof table.
- ☐ Cleanup, restored ARP and no remaining processes.

Student Report Template

9. Executive summary and authorization.
10. Lab network diagram.
11. Baseline evidence and hashes.
12. Controlled simulation method.
13. DNS anomaly timeline.
14. ARP correlation.
15. Victim connection correlation.
16. Alternative explanations and confidence level.
17. Cleanup and defensive recommendations.
18. Appendix.

Submission Requirements

- One PDF report named SBT-DF203-Lab8_RegNo_FullName.pdf.
- The original or generated PCAP/PCAPNG evidence file and its SHA-256 hash text file.
- A reports folder containing exported CSV/TXT command output and any recovered objects.
- Screenshots must be readable, numbered and referenced in the report narrative.
- Compress the report and evidence outputs into one ZIP named with your registration number and lab number.
- Do not submit passwords or decoded credentials in public discussion channels; include them only in the protected assessment submission where required.

Marking Rubric (100 Marks)

Assessment Area	Marks	What the Instructor Will Check
Safety and preparation	20	Reserved domain, no credential capture, host-only network and baseline recorded.
DNS analysis	25	Correct query/response, transaction, responder, answer and TTL analysis.
ARP and connection correlation	20	Links DNS anomaly to ARP and subsequent traffic.
Forensic interpretation	15	Considers competing explanations and states confidence.
Cleanup and defenses	10	Restored environment and appropriate controls.
Report quality	10	Professional timeline and screenshots.

Instructor Quick Answer Guide

Checkpoint	Expected Observation
Core indicator	DNS answer points to an unauthorized replacement IP.
Strong packet indicator	Resolver IP appears with an unexpected Ethernet source MAC.
Race indicator	Two conflicting responses share one transaction/query.
ARP correlation	Gateway/victim mapping changes to analyst MAC.
Impact evidence	Victim connects to the forged answer.
HTTPS control	Certificate validation limits successful impersonation of real sites.

Command Reference Summary

Command	Purpose
dns.flags.response	Query/response indicator.
dns.id	Match transaction.
dns.a	IPv4 answer.
dns.resp.ttl	Response TTL.
arp.src.proto_ipv4 / hw_mac	ARP IP-to-MAC claim.
http.host	Requested HTTP host.

Academic Integrity and Authorization Statement

Student Declaration

I confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.